PLACIDWAY KNOWLEDGE CHATBOT

Technical Project Documentation

n8n + Supabase Vector Store + PostgreSQL Chat Memory + Chatwoot + WhatsApp

Project type	Knowledge-based conversational support automation
Workflow	1-PlacidWay Knowledge Chatbot (N8N+SUPABASE+CHATWOOT)
Document version	1.0
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Purpose

This document describes the supplied n8n workflow, its connected services, the message-processing path, the visible test evidence, and the key operational checks for maintaining the chatbot.

Confidentiality note: The exported workflow contains connection references and a Chatwoot API access token. Treat the original JSON as sensitive. Rotate any exposed token before sharing the export publicly.

1. Executive Summary

The project connects incoming Chatwoot conversation events to an n8n AI workflow. The workflow extracts conversation and sender fields, routes messages by message type, uses an OpenAI chat model with PostgreSQL-backed chat memory, and exposes a Supabase vector store as a retrieval tool for answering questions from indexed knowledge documents. The generated answer is posted back to the relevant Chatwoot conversation.

The supplied screenshots show successful n8n executions, rows in the Supabase documents table containing text/metadata/embedding vectors, a WhatsApp test conversation, and a Chatwoot inbox with a human takeover option. These are evidence of observed test activity, not a guarantee of every production scenario.

2. Architecture and Message Flow

Step	Component	Role
1	Chatwoot webhook → n8n Webhook	Receives POST events at the workflow path named chatwoot.
2	Edit Fields	Maps account ID, conversation ID, message type, sender name, phone number, and message content into consistent fields.
3	Switch	Routes the message based on message_type; the configured incoming branch continues to the AI Agent.
4	AI Agent + OpenAI Chat Model	Generates a concise answer using the defined response rules and the connected model (configured as gpt-5-mini).
5	Supabase Vector Store + OpenAI Embeddings	Retrieves relevant chunks from the documents table as an AI tool; embeddings support vector similarity retrieval.
6	Postgres Chat Memory	Uses the sender phone number as the custom session key for conversation memory.
7	Send Message (HTTP Request)	Posts the AI output as an outgoing, non-private text message to the matching Chatwoot conversation.
Parallel	Execute a SQL query	Upserts the contact phone number and last-message timestamp into whatsapp_users.

Simplified flow

Customer sends WhatsApp message → Chatwoot receives it → webhook triggers n8n → fields are mapped → incoming-message route → AI Agent retrieves relevant knowledge and uses conversation memory → answer is sent to Chatwoot → Chatwoot delivers it through the connected channel.

3. Main Components

n8n

Orchestrates the workflow. The imported workflow is marked active in the supplied JSON. The screenshot shows successful execution entries, including runs completed in milliseconds and one run lasting several seconds.

Chatwoot

Provides the conversation inbox and agent interface. The screenshot indicates the conversation can be handled by the automation and offers a “Take over” option for human intervention.

Supabase Vector Store

The workflow uses the table named documents in retrieve-as-tool mode. The supplied Supabase screenshot shows 57 records and columns for id, content, metadata, and embedding vector. This is consistent with indexed knowledge chunks being available for retrieval.

OpenAI

The workflow configures an OpenAI chat model named gpt-5-mini and an OpenAI embeddings node. Credentials are referenced by n8n credential entries rather than documented as values here.

PostgreSQL

Postgres Chat Memory uses the sender phone number as its session key. A separate SQL node inserts the WhatsApp source ID into whatsapp_users and updates last_message_at on conflict.

4. AI Response Rules

The AI Agent system message in the supplied workflow specifies the following behavior:

- Answer in 1–3 short sentences or at most 3 bullets, targeting a maximum of about 60 words before the source line.
- Lead with the direct answer and avoid introductory filler, emojis, and closing questions.
- Answer only what the user asked; generally provide one price unless the user asks for multiple prices or a comparison.
- Include a source line in the format: “Source: title - URL”. Omit a link if no source link is available.
- Give longer answers only when the user requests details, a full list of inclusions, or a comparison.

Important scope note

The screenshots demonstrate example medical-treatment information and a cautious response to a question about safety with Type 2 diabetes. The chatbot should not be treated as a medical professional. Eligibility, diagnosis, treatment decisions, and safety questions should be referred to the provider’s qualified medical team; the bot should not guarantee outcomes or independently confirm that a treatment is safe.

5. Data and Database Behavior

Knowledge data

The vector-store node queries the documents table. The screenshot shows content snippets and metadata including location/page-line information alongside vector embeddings. The provided workflow JSON does not include the original source documents or the ingestion workflow, so this document does not claim how the 57 records were created.

Conversation memory

The Postgres Chat Memory node is configured with a custom session key based on phonenumber. Verify the memory table and retention policy in the connected database before production use.

Contact activity tracking

The SQL node uses an upsert pattern for whatsapp_users, recording the source ID and refreshing the last-message timestamp. Validate the source ID format and SQL parameter handling with real webhook payloads.

6. Testing Evidence

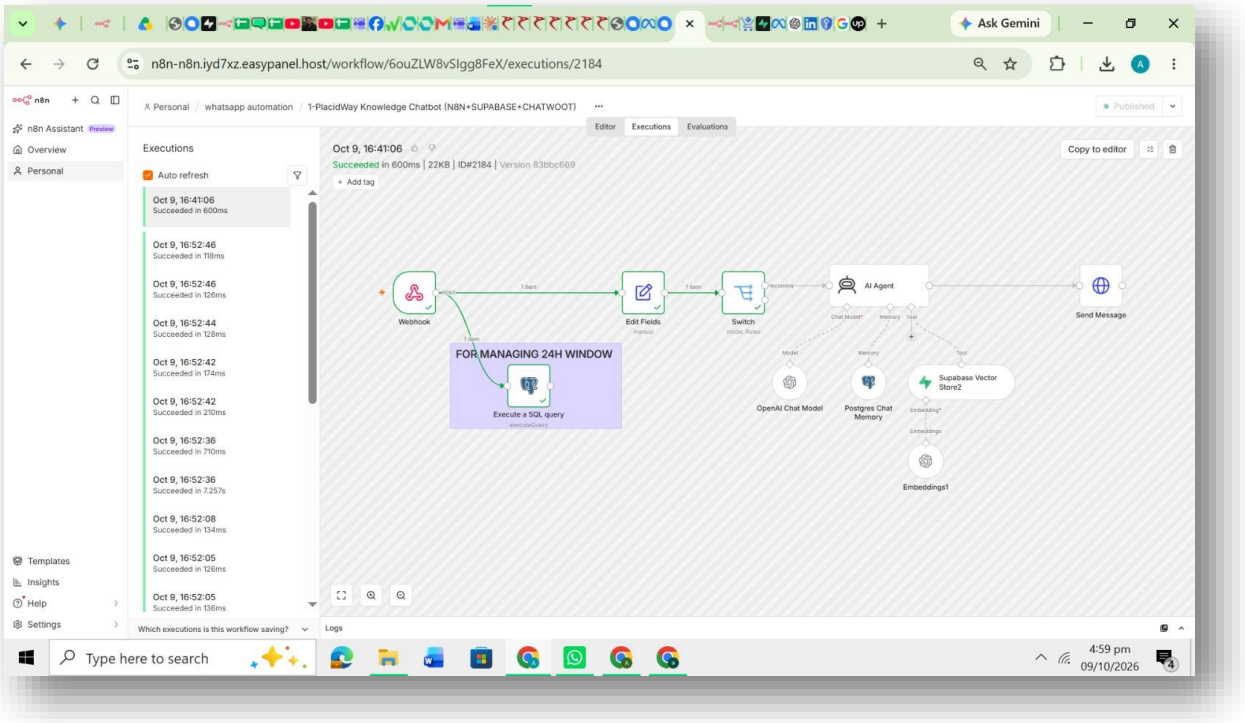


Figure 1. n8n workflow execution history showing successful runs.

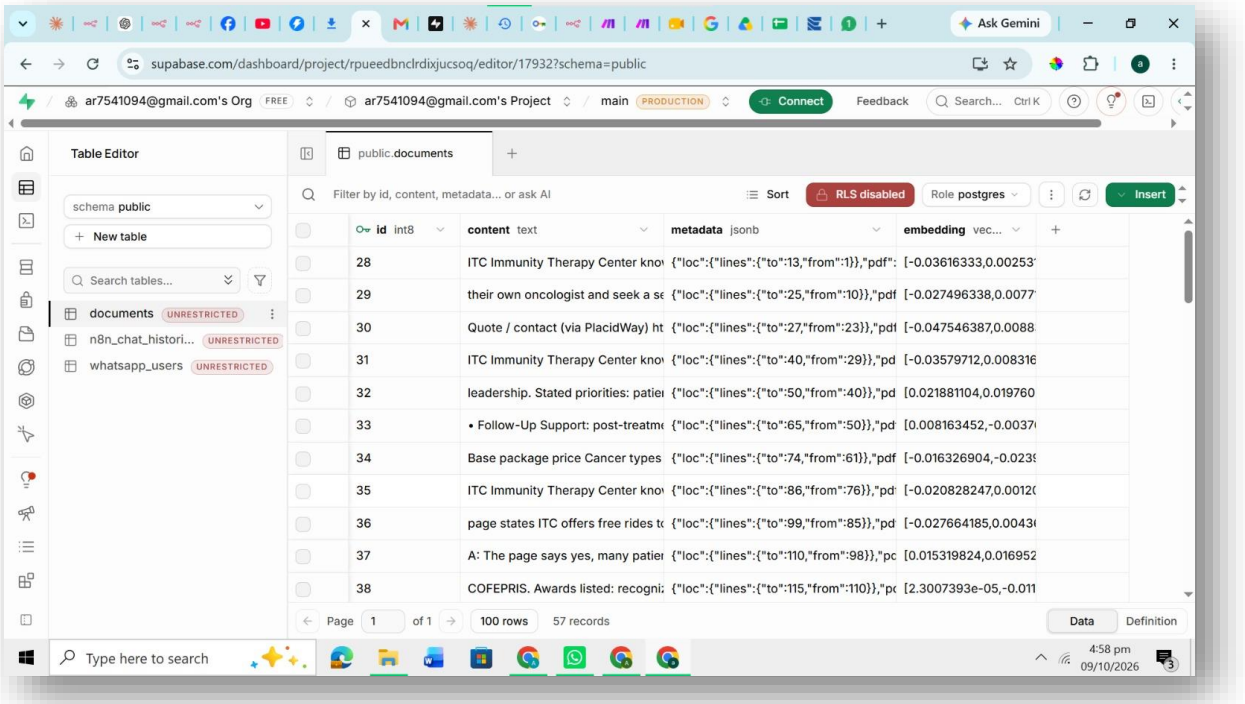


Figure 2. Supabase documents table with content, metadata, and embedding vectors.

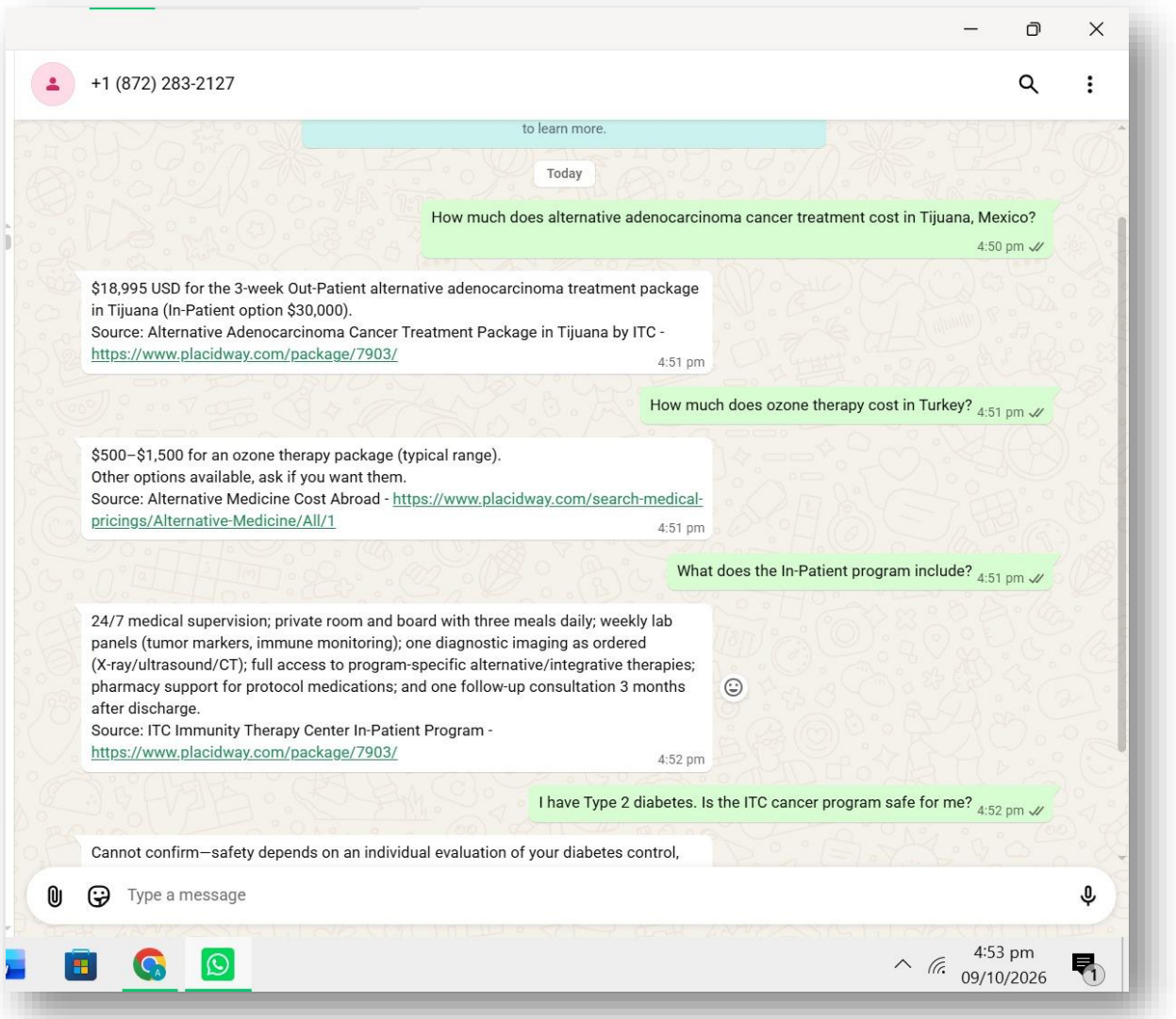


Figure 3. WhatsApp conversation demonstrating answers about treatment pricing and program inclusions.

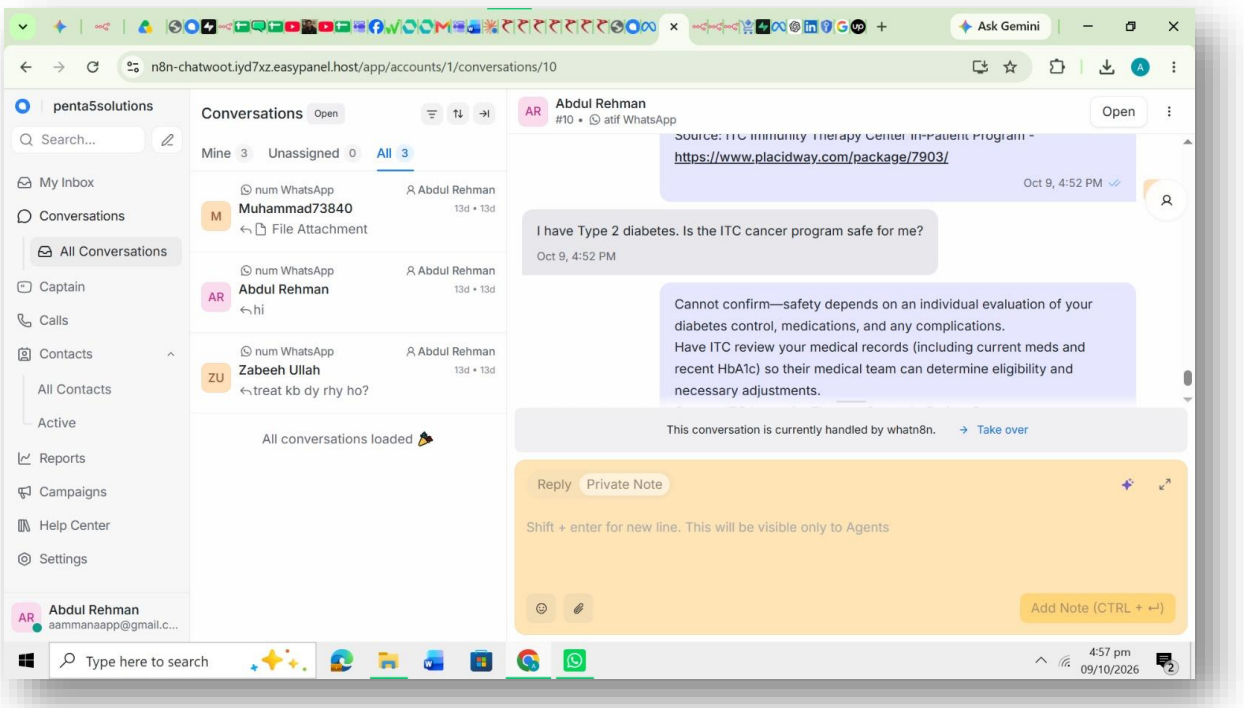


Figure 4. Chatwoot conversation inbox showing AI-handled messages and human takeover option.

Observed test examples

The WhatsApp screenshot includes questions about alternative adenocarcinoma treatment pricing in Tijuana, ozone therapy pricing in Turkey, and what an in-patient program includes. Another exchange asks whether the program is safe for a person with Type 2 diabetes; the shown answer does not confirm safety and recommends individual review by the provider's medical team.

7. Setup and Configuration Checklist

- ☐ Import the supplied workflow JSON into the intended n8n instance and review all node connections.
- ☐ Reconnect or confirm n8n credentials for OpenAI, Supabase, and PostgreSQL. Do not rely on credential IDs from another n8n instance.
- ☐ Configure the Chatwoot webhook to send the correct conversation events to the n8n production webhook URL.
- ☐ Confirm the Chatwoot account ID and conversation ID fields match the incoming payload structure.
- ☐ Confirm the Supabase documents table schema and vector dimensions are compatible with the selected embedding model and vector-store configuration.
- ☐ Confirm the PostgreSQL chat-memory database/table setup and that phone numbers are normalized consistently.
- ☐ Test incoming messages, outgoing/self-authored events, empty content, attachments, unknown message types, and duplicate webhook deliveries.
- ☐ Verify the WhatsApp/Chatwoot channel's message-window and template rules before sending replies outside the permitted customer-service window.
- ☐ Review logs and execution data for failed nodes, unexpected payloads, latency, and model/retrieval errors.
- ☐ Document a human handoff process for clinical, billing, complaints, or low-confidence questions.

8. Security, Privacy, and Reliability

- Rotate exposed credentials. The supplied workflow export contains a literal Chatwoot API access token. Revoke/rotate it immediately if the export has been shared outside trusted operators, and store the replacement in n8n credentials or a secure secret store.
- Protect health-related conversations. Restrict access to Chatwoot, n8n executions, database records, and backups. Define retention and deletion rules; avoid logging unnecessary personal or medical details.
- Harden SQL handling. The SQL query interpolates a webhook value into SQL text. Prefer parameterized queries and validate input to reduce SQL-injection risk.
- Check event filtering. The Switch node has configured incoming and outcoming rules, but only the incoming output is connected onward. Confirm outgoing events are intentionally ignored and that the actual Chatwoot payload uses the expected spelling/value.
- Verify response window logic. A sticky note mentions management of the 24-hour window, but the provided workflow JSON does not show a connected node implementing a complete window check. Do not assume the note itself enforces the rule.
- Validate source attribution. Ensure the AI's source title and URL correspond to retrieved material. Test unsupported questions and retrieval misses.
- Back up critical data. Back up the workflow export, knowledge source files, Supabase database, and relevant configuration securely.

9. Known Limitations / Items to Verify

- The provided export shows the conversational workflow, not the full knowledge-ingestion pipeline.
- No explicit confidence threshold, fallback response, retry strategy, or error workflow is visible in the supplied JSON.
- The SQL tracking node runs in parallel from the webhook, so verify that a SQL failure cannot affect the main response path and that failures are monitored.
- The exact Chatwoot event subscription, WhatsApp channel configuration, and provider-side 24-hour enforcement are not fully established by the export.
- Screenshots represent the test state at capture time; they do not prove ongoing uptime, accuracy, security compliance, or complete coverage.

10. Acceptance Test Plan

Test	Expected result
Simple knowledge question	A concise answer grounded in a relevant retrieved document, with source title/URL.
Price question	One directly relevant price and its source; no unsupported price claims.
Medical safety question	No definitive safety guarantee; direct user to qualified clinical review.
Follow-up question	Memory is scoped to the correct sender/session and does not mix users.
Outgoing Chatwoot event	Does not trigger an unwanted duplicate AI reply.
Human takeover	Human agent can take over and automation does not interfere with the manual conversation.
Missing/invalid payload	Execution fails safely or follows a defined fallback; error is visible to operators.
Message-window boundary	Permitted reply method is used according to the active WhatsApp/Meta rules.

11. Workflow Inventory

The supplied workflow contains these nodes: Webhook, Edit Fields, AI Agent, Switch, OpenAI Chat Model, Send Message, Postgres Chat Memory, Execute a SQL query, Supabase Vector Store2, Embeddings1, Sticky Note.

12. Handover Summary

This implementation demonstrates a WhatsApp-facing knowledge chatbot routed through Chatwoot and n8n. Supabase provides retrieval over indexed knowledge chunks, PostgreSQL provides chat memory and contact activity tracking, and the AI response is sent back to the originating Chatwoot conversation. Before public production handover, prioritize credential rotation, secure data access, parameterized SQL, message-type filtering, 24-hour-window validation, error handling, and end-to-end acceptance tests.